Algebra 2

Slope, Revenue & Area Drill — Functions & Logarithms

Name: ______________ Block: ___________ Date: ___________

~35 minutes · 46 points total. Calculator allowed on Q2. Q4–Q7 are no calculator. Show all work.

Q1 — Interpreting Slope on a Mountain Road

8 pts

1.

A mountain road drops **42 feet** in elevation for every **700 feet** of horizontal distance.

(a) What is the percent grade of this road? Show your calculation.

Grade = __________ %

(b) Write a linear equation for vertical drop V (feet) as a function of horizontal distance H (feet).

$V =$ ________________________

(c) A driver starts at the top. After driving some distance, the altimeter shows the car is **84 feet lower** than the start. How far has the car traveled horizontally?

$H =$ ____________ feet

(d) The base of the hill is **2,500 feet** horizontally from the top. What is the total height of the hill?

Height = ____________ feet

Q2 — Revenue as a Function (Calculator)

12 pts

2.

A streaming service charges $(150 - 7x)$ dollars per subscription, where x is the number of subscriptions sold (in thousands). Revenue is price times quantity, so $R(x) = x(150 - 7x)$, measured in thousands of dollars.

(a) Write $R(x)$ in standard form $ax^2 + bx + c$. Identify a, b, and c.

$R(x) =$ ______________________

$a =$ ________ $b =$ ________ $c =$ ________

(b) What values of x make $R(x) = 0$? What do those values mean in context?

$x =$ ______________________

Meaning: ____

(c) Find the value of x that maximizes revenue. What is the maximum revenue?

$x =$ ________________ thousand subscriptions

Max $R =$ ________________ thousand dollars

(d) The company needs $R(x) = 700$ (thousand dollars). Solve for x. There are two solutions — what does each one mean for the business?

$x_1 \approx$ ______________ thousand $x_2 \approx$ ______________ thousand

Meaning: ____

Q3 — Triangle Under a Line

14 pts

3.

The line $f(x) = 48 - 6x$ is drawn on a coordinate plane. A right triangle is formed with its base along the x-axis from 0 to x, and its height going straight up from $(x, 0)$ to the point $(x,\ f(x))$ on the line.

(a) Write the area function $A(x)$. Simplify completely.

$A(x) =$ ______________________________

(b) What is the domain of $A(x)$? Explain why x cannot go higher than its upper bound.

Domain: ____

(c) Complete the table.

x	1	2	3	4	5	6	7
$A(x)$							

What pattern do you notice in the table?

(d) Find the value of x that maximizes the area. What is the maximum area?

$x =$ __________ Max area = ____________ square units

(e) Find all values of x where $A(x) = 45$. Describe the two triangles.

$x =$ __________ and __________

(f) Find all values of x where $A(x) = 36$. For each solution, write the base and height of the triangle and explain why the two triangles have the same area even though they look different.

$x =$ __________ and __________

Explanation: ____

Q4 — Evaluate $\log_{27}(9)$ (No Calculator)

3 pts

4. Evaluate: $\log_{27}(9)$

$\log_{27}(9) =$ ____________

Q5 — Evaluate $\log_{32}(8)$ (No Calculator)

3 pts

5. Evaluate: $\log_{32}(8)$

$\log_{32}(8) =$ ____________

Q6 — Evaluate $\log_{16}(32)$ (No Calculator)

3 pts

6. Evaluate: $\log_{16}(32)$

$\log_{16}(32) =$ ____________

Q7 — Evaluate $\log_{125}(25)$ (No Calculator)

3 pts

7. Evaluate: $\log_{125}(25)$

$\log_{125}(25) =$ ______________